National University

O f C o m p u t e r & E m e r g i n g S c i e n c e s F a i s a l a b a d - C h i n i o t C a m p u s

CS 1005 : Discrete Structures

Assignment 3

Instructor : Ms Mahzaib Younas

SECTION BCS (C)

Q No 01: Determine whether these relations are reflexive, symmetric, irreflexive, antisymmetric and/or transitive. (Marks 15)

1. R1= {(a,b) | a and b have no letters in common}
2. R2 = {(a,b) | a and b are not the same length}
3. R3 = {(a,b) | a is longer than b}

QNO02: Consider the set A ={1,2,3} with the relations R = { (1,1) , (1,3) , (2,1), (2,2), (3,1)} and S = {(1,2), (2,1), (2,2), (3,3)}. Calculate the composition R.S in matrix form. Finally convert the matrix into listing method. (Marks 5)

QNO03: Let R be a reflexive relation on set A. Show that $R \subseteq R^2$. (Marks 5)

QNO04: Let R be a relation on the set of real numbers such that a R b if and only if a-b is an integer. Is R an equivalent relation? (Marks 5)

QNO05: Assume that R is a relation on a set A. Prove or dis-prove each statement. (Marks 15)

1. If R is reflexive, then R−1 is reflexive.
2. If R is symmetric, then R−1 is symmetric.
3. If R is transitive, then R−1 is transitive.

QNO06: Let A = {1, 2, 3} and B = {1, 2, 3, 4}. The relations R1 = {(1, 1), (2, 2), (3, 3)} and R2 = {(1, 1), (1, 2), (1, 3), (1, 4)} can be combined to R1 ∪ R2, R1 ∩ R2, R1 ⊕ R2, R1 − R2, and R2 − R1? (Marks 5)

QNO07: Let A and B be the set of all students and the set of all courses at a school, respectively. Suppose that R1 consists of all ordered pairs (a, b), where a is a student who has taken course b, and R2 consists of all ordered pairs(a, b), where a is a student who requires course b to graduate. What are the relations R1 ∪ R2, R1 ∩ R2, R1 ⊕ R2, R1 − R2, and R2 − R1? (Marks 5)

QNO08: Congruence modulo m. Let m be an integer with m > 1. Show that the relation (Marks 5)

$$\mathbf{R = \{(a,b) \mid a \equiv b \pmod{m}\}}$$

QNO09: Let X be the set of all nonempty subsets of {1, 2, 3}. Then X = {{1},{2},{3},{1, 2},{1, 3},{2, 3},{1, 2, 3}} Define a relation R on X as follows: For all A and B in X, A R B ⇔ the least element of A equals the least element of B. Prove that R is an equivalence relation on X. (Marks 5)

QNO10: Suppose that S = {a, b, c, d, e} and R is a relation on S such that aRb, bRc and dRe. List all of the following that must be true if R is (a) reflexive (but not symmetric or transitive), (b) symmetric (but not reflexive or transitive), (c) transitive (but not reflexive or symmetric), and (d) an equivalence relation. (Marks 10)

1. cRb
2. cRc
3. aRc
4. bRa
5. aRd
6. eRa
7. eRd
8. cRa